

Discrete Math — Monday Test

Theoretical / Conceptual Questions (Rosen’s own end-of-chapter Review Questions)

These are Rosen’s official *Review Questions* from the end of Chapters 1 and 5 — conceptual, “explain/define/describe” style questions, as opposed to the computational exercises in the other two PDFs. They’re exactly the kind of thing a professor asks to check you understand the *ideas*, not just the mechanics. Numbered T1–T18 to keep them separate from Q1–Q38 in the other PDFs. Organized by the same day schedule.

Tuesday — after §1.1 & §1.3

T1. a) Define the negation of a proposition.

b) What is the negation of “This is a boring course”?

T2. a) Define (using truth tables) the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions p and q .

b) What are the disjunction, conjunction, exclusive or, conditional, and biconditional of the propositions “I’ll go to the movies tonight” and “I’ll finish my discrete mathematics homework”?

T3. a) Describe at least five different ways to write the conditional statement $p \rightarrow q$ in English.

b) Define the converse and contrapositive of a conditional statement.

c) State the converse and the contrapositive of the conditional statement “If it is sunny tomorrow, then I will go for a walk in the woods.”

T4. a) What does it mean for two propositions to be logically equivalent?

b) Describe the different ways to show that two compound propositions are logically equivalent.

c) Show in at least two different ways that the compound propositions $\neg p \vee (r \rightarrow \neg q)$ and $\neg p \vee \neg q \vee \neg r$ are equivalent.

T5. a) Given a truth table, explain how to use disjunctive normal form to construct a compound proposition with this truth table.

b) Explain why part (a) shows that the operators $\wedge$, $\vee$, and $\neg$ are functionally complete.

c) Is there an operator such that the set containing just this operator is functionally complete?

Now do: T1, T2, T3, T4, T5

Wednesday — after §1.4

T6. What are the universal and existential quantifications of a predicate $P(x)$? What are their negations?

Now do: T6

Only one question this section — spend the rest of the day's time on Q9–Q13 from the Practice Problems PDF instead.

Thursday — after §1.5 & start §1.6

T7. a) What is the difference between the quantification $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$, where $P(x, y)$ is a predicate?

b) Give an example of a predicate $P(x, y)$ such that $\exists x \forall y P(x, y)$ and $\forall y \exists x P(x, y)$ have different truth values.

T8. Describe what is meant by a valid argument in propositional logic and show that the argument “If the earth is flat, then you can sail off the edge of the earth,” “You cannot sail off the edge of the earth,” therefore, “The earth is not flat” is a valid argument.

Now do: T7, T8

Friday — after §1.6 (finish)

T9. Use rules of inference to show that if the premises “All zebras have stripes” and “Mark is a zebra” are true, then the conclusion “Mark has stripes” is true.

Now do: T9

Also light today — put the remaining time into Q20–Q22 from the Practice Problems PDF, which cover the same ground with more reps.

Saturday — after §1.7 & §1.8

T10. a) Describe what is meant by a direct proof, a proof by contraposition, and a proof by contradiction of a conditional statement $p \rightarrow q$.

b) Give a direct proof, a proof by contraposition, and a proof by contradiction of the statement: “If n is even, then $n + 4$ is even.”

T11. a) Describe a way to prove the biconditional $p \leftrightarrow q$.

b) Prove the statement: “The integer $3n + 2$ is odd if and only if the integer $9n + 5$ is even, where n is an integer.”

T12. To prove that the statements p_1, p_2, p_3 , and p_4 are equivalent, is it sufficient to show that the conditional statements $p_4 \rightarrow p_2$, $p_3 \rightarrow p_1$, and $p_1 \rightarrow p_2$ are valid? If not, provide another collection of conditional statements that can be used to show that the four statements are equivalent.

T13. a) Suppose that a statement of the form $\forall x P(x)$ is false. How can this be proved?

b) Show that the statement “For every positive integer n , $n^2 \geq 2^n$ ” is false.

T14. What is the difference between a constructive and nonconstructive existence proof? Give an example of each.

T15. What are the elements of a proof that there is a unique element x such that $P(x)$, where $P(x)$ is a propositional function?

T16. Explain how a proof by cases can be used to prove a result about absolute values, such as the fact that $|xy| = |x||y|$ for all real numbers x and y .

Now do: T10, T11, T12, T13, T14, T15, T16

This is the heaviest theory day — these seven questions map almost one-to-one onto what a professor would ask to check you actually understand proof strategy, not just execute it. Don't skip T12 and T13; they're the two most commonly missed.

Sunday — after §5.1

T17. a) Can you use the principle of mathematical induction to find a formula for the sum of the first n terms of a sequence?
b) Can you use the principle of mathematical induction to determine whether a given formula for the sum of the first n terms of a sequence is correct?
c) Find a formula for the sum of the first n even positive integers, and prove it using mathematical induction.

T18. a) For which positive integers n is $11n + 17 \leq 2^n$?
b) Prove the conjecture you made in part (a) using mathematical induction.

Now do: T17, T18

T18 is a genuinely good exam-style question — it makes you first find the pattern (try small n) before proving it, exactly like the "conjecture then prove" problems you've already drilled.